

AMENDMENT NO. 4 JUNE 2022

TO

IS 14664 : 2010 AUTOMOTIVE VEHICLES — PERFORMANCE REQUIREMENTS AND TESTING PROCEDURE FOR BRAKING SYSTEM OF TWO AND THREE WHEELED MOTOR VEHICLES

(First Revision)

(Page 1, clause 2) — Insert the following at the end of the clause:

‘ISO 2575 : 2010 Road vehicles — Symbols for controls, indicators and tell-tales’

[Page 1, clause 3.30 (see also Amendment No. 1)] — Insert the following new clause:

‘3.31 Disable the Antilock Brake System — To put the system into a state where it will no longer fulfill the technical requirements given in 5.9.’

[Page 4, clause 4.1.15 (see also Amendment No. 1)] — Substitute the following for the existing clause:

‘4.1.15 If a vehicle is equipped with the means to indicate emergency braking, activation and deactivation of the emergency braking signal shall only be generated by the application of any service braking system when the conditions in clauses 4.1.15.1 and 4.1.15.2 are fulfilled.

4.1.15.1 The signal shall not be activated when the vehicle deceleration is below 6 m/s², but it may be generated at any deceleration at or above this value, the actual value being defined by the vehicle manufacturer. The signal shall be deactivated at the latest when the deceleration has fallen below 2.5 m/s².

4.1.15.2 The signal may be activated at a speed above 50 km/h when the antilock system is fully cycling [as defined in 5.9.1(c)] and deceleration is

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at least 2.5 m/s². The signal shall be deactivated when the antilock system is no longer fully cycling.

4.1.15.3 The signal may be generated from a prediction of the vehicle deceleration resulting from the braking demand respecting the activation and deactivation thresholds defined in **4.1.15.1.**

[Page 4, clause **4.1.15.2** (see also Amendment No. 1)] — Insert the following new clauses:

4.1.16 A means to disable the antilock brake system is allowed, if a vehicle is fitted with a riding mode selector allowing an “off road” or “all terrain” mode, and the following provisions are met:

- a) Disabling of the antilock brake system shall only be allowed when the riding mode selector is in the “off road” or “all terrain” mode.
- b) The vehicle is stationary.
- c) The disablement of the antilock brake system function shall be the result of a deliberate action by the rider according to one of the following methods:
 - 1) Simultaneous actuation of the antilock brake system disable switch and a service brake system control (that is brake lever or pedal); or
 - 2) The actuation of the antilock brake system disable switch for a minimum of 2s; or
 - 3) The progression through at least two successive steps or levels of actuation of a control (for example, rotating knob, a touch panel or a menu option selector).
- d) The antilock brake system function shall be automatically enabled when exiting from the “off road” or “all terrain” ride mode, or after each start-up of the vehicle.
- e) When disabled, the antilock brake system function shall be indicated by the activation of a yellow or amber tell-tale according to one of the following methods until the Antilock Brake System (ABS) is fully functional or operating again :

- 1) The following symbol as specified in **B.18** in ISO 2575 :



; or

- 2) The following symbol as specified in **B.05** of ISO 2575:



; or

- 3) With the word “OFF” as follows, according to **Y.01** in ISO 2575, whereby the tell tales are adjacent to each other:

OFF

; or

- 4) The text “ABS OFF” or “ABS not available”; or
- 5) The warning lamp referred in **4.1.13**, continuously activated (that is lit or flashing).
- f) Enabling of a functional stage which complies with the antilock brake system requirements in **5.9** shall be possible through the single actuation of a control (for example, simple press of a button or switch) initiating ABS system start-up procedure.
- g) The manufacturer shall not make available to consumers, additional means of disabling ABS other than in compliance with **4.1.16 (a)** to **4.1.16 (f)**. This does not apply to what is required to service the ABS (for example, electrical connectors).

4.1.17 A vehicle fitted with an ABS system active on both axles may be fitted with a rider selectable mode to deactivate the ABS function on the rear axle. When the ABS function is deactivated on the rear axle this shall be indicated by a yellow or amber tell-tale or check control messages (pop-up messages in the instrument panel) according to one of the following methods until the ABS is fully functional or operating on both axles again:

- a) The following symbol as specified in **B.18** in ISO 2575:



- with the word “REAR” adjacent to it; or

- b) The following symbol as specified in **B.18** in ISO 2575:



- with a symbol of the vehicle adjacent to it with an arrow pointing to the rear axle; or

- c) The following symbol as specified in **B.05** of ISO 2575:



- with the word “REAR OFF” adjacent to it; or

- d) The text “REAR ABS OFF” or REAR ABS not available”; or

- e) The warning lamp referred to in **4.1.13** continuously flashing. If the disablement of the ABS system is also indicated by the flashing of this warning lamp as specified in **4.1.16 (e) (4)**, the frequency of the flashing for indicating the deactivation of the ABS system on one axle shall be different from the frequency of the flashing to indicate the disablement of the ABS system.

NOTE — The switching from dual-channel ABS function to single-channel ABS function is considered as switching from one ABS functional stage to another ABS functional stage and therefore the requirements of **4.1.16** do not apply as long as the system complies with the requirement of **5.9**.

4.1.18 If fitted, the provisions in **4.1.16** and **4.1.17** shall be applicable for category 3-1 and 3-3 vehicles manufactured on and after 1 September 2023. However, at the request of the applicant, a vehicle may be tested for compliance as per **4.1.16** and **4.1.17** before 1 September 2023. In case of vehicle models already approved for similar provision as specified in **4.1.16** and **4.1.17** before 1 September 2023, approvals shall be extended without any further testing.’

[Page 13, clause **5.9.1 (a)**] — Substitute the following for the existing clause:

- ‘a) The tests are only applicable to the ABS, if fitted and enabled on vehicle categories 3-1 and 3-3.’

[Page 13, clause **5.9.1 (d)**] — Insert the following new clause:

- ‘(e) Vehicles with rider selectable ABS modes (for example, a dual channel ABS system whereby the ABS on the rear axle can be disabled) shall comply with the technical requirements of **5.9** in all modes where ABS is enabled.’